Week 6

Singular Value Decomposition

$$A = USV^T$$

where

- A is $m \times n$ matrix
- U is $m \times m$ matrix
- S is $m \times n$ matrix: diagonal matrix
- V^T is $n \times n$ matrix

- U and V^T are orthogonal i.e. $U^TU = I$ and $V^TV = I$.
- $AA^T = UDD^TU^T$
- $A^TA = VD^TDV^T$

Finding SVD

Step 1: Find eigenvalues and eigenvectors of A^TA. Using eigenvectors, form an orthonormal basis of $\mathbb{R}^2$.

Step 2: Use $\sigma_1 y_1 = Ax_1$ and $\sigma_2 y_2 = Ax_2$ (x_1 and x_2 is the basis from Step 1) to find y_1 and y_2.

Step 3: $U = \begin{bmatrix} | & | \\ y_1 & y_2 \\ | & | \end{bmatrix}$ $V = \begin{bmatrix} | & | \\ x_1 & x_2 \\ | & | \end{bmatrix}$ and $S = \begin{bmatrix} \sqrt{\sigma_1} & 0 \\ 0 & \sqrt{\sigma_2} \end{bmatrix}$

Positive Definiteness

💡 Every quadratic function of the form $ax^2 + 2bxy + cy^2$ has a stationary point at $(0, 0)$.

Positive Definite

A function f that vanishes at $(0, 0)$ and is strictly positive at all other points and is denoted by $f > 0$.

What condition on a, b, c ensure $f(x, y) = ax^2 + 2bxy + cy^2$ is positive definite?

Necessary Conditions

1. $a > 0$
2. $c > 0$
3. $ac > b^2$

Positive Semi-definite

if $ac = b^2$ and $a > 0$

Negative Semi-definite

if $ac = b^2$ and $a < 0$

Saddle point at $(0, 0)$

if $ac < b^2$

Connection to Linear Algebra

$$ax^2 + 2bxy + cy^2 = \begin{bmatrix} x & y \end{bmatrix} \begin{bmatrix} a & b \\ b & c \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

$$\text{Let } v = \begin{bmatrix} x \\ y \end{bmatrix} \text{ and } A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$\text{Then, } ax^2 + 2bxy + cy^2 = v^T Av$$

Positive Definite Matrices

$A = \begin{bmatrix} a & b \\ b & c \end{bmatrix}$ is positive definite if $a > 0, ac - b^2 > 0$.

- If $a > 0$ and $ac - b^2 > 0$, then both eigenvalues of A are positive.

A real symmetric $n \times n$ matrix is positive definite if

$v^T Av > 0 \; \forall \; v \in \mathbb{R}^n, v \neq 0$ and all eigenvalues/pivots are positive.